

MAMADOU B. BAH Denver, CO | 718-650-7411 | mamadoubah2277@gmail.com
Digital Portfolio: [<https://mamadoubbah.github.io/>] |

Technical Educator | Software Developer | Data & Educational Systems Specialist

PROFESSIONAL PROFILE Versatile Technical Lead with over 10 years of experience in **Cross-Functional Leadership, Data Analysis, Technology & Software Education, Social Work & Counseling**. Expert in designing educational curriculum that prioritizes accessibility, practicality, and student understanding through project application. Proven track record of collaborating in multi-disciplinary teams (doctors, psychologists, therapists, IT personnel etc.) in high-pressure environments. Seeking a Remote or Hybrid role in Software Development, Management, AI Training, or EdTech.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java, JavaScript, HTML/CSS, R/RStudio.
- **Specialized Tech:** AI Tutoring Logic (RAG), MIDI/Audio Engineering, Data Visualization, BigQuery
- **Operations:** Cost-Benefit Analysis, Curriculum design, Ethical Implementations, Special needs adaptability, Cross-Functional Team Management.

PORTFOLIO & TECHNICAL PROOF OF WORK

- **FinEdge Web App:** Developed a full-stack application designed for ease of use in budget tracking, building demo stock portfolios, and researching stock symbols.
- **BlackCode (AI Educational Assistant):** Developed an offline RAG-based AI tutor designed to grow alongside students and maintain persistent user context, as well as discern user intent goals, all while providing factual responses.
- **Environmental Data Analysis:** Made use of NOAA meteorological data to analyze and conduct a case study of data centers and their influence on severe weather effects.
- **Digital Advent Curriculum:** A fully designed 3 year curriculum meant to introduce the foundations of programming and software use in tandem with the creative arts. Later years cover the more advanced fundamentals of programming and cybersecurity, alongside the inclusion of Artificial Intelligence.

PROFESSIONAL EXPERIENCE

Head of Middle School Computer Science | *Denver, CO* | 2023 – 2026

- **Curriculum Designer:** Engineered a multi-level CS curriculum integrating Creative Arts ,Technology and software use to both keep students engaged and prepared for the future. Implemented "scaffolding" logic for students with special needs and complications to be able to still actively participate and learn.
- **Project Based Learning:** Designed projects, tests, and assignments that focus on skill acquisition over simple memorization. Covered the subject areas of web development,

personal finance, Audio Engineering, Film production, and game development in the context of technology use for future student career paths.

- **Hardware & Fiscal Oversight:** Conducted school-wide cost-benefit analysis for infrastructure; helped convince administration to invest in technological upgrades to serve hundreds of students, providing them with up to date desktops with modern RAM capabilities. The upgrade allowed for more advanced projects to be performed such as introducing students to data science. We were able to cover upgrades for over 150 additional students in this way.
- **Data Leadership:** Gathered student grades as well as student survey results to perform Data Analysis, identifying learning gaps and developing pivot instructional strategies in real-time. Also used student data to modify curriculum to better engage students overall, improve accessibility and reduce language-related learning barriers.

AP Computer Science / Game Development Professor | *Charter School* | 2022 – 2023

- **Multi-Stack Instruction:** Led three advanced courses (AP CS, General CS, Game Dev) covering Java, SQL, and Web Development, and game design.
- **Pedagogical Modeling:** Translated complex programming and data science concepts into visual and verbal models that improved student comprehension.

New York City Department of Education | *New York, NY* | 2020 – 2021 Lead Special Education Teacher (D75)

- **Crisis & Team Leadership:** Coordinated with a multidisciplinary team of doctors, therapists, and psychologists to provide holistic care for students with severe disabilities.
- **Systemic Problem Solving:** Authored 100+ Individualized Educational Plans (IEPs), utilizing data-driven benchmarks to track and ensure the autonomy of students with autism and non-verbal conditions.
- **Intensive Curriculum Scaffolding:** Adjusted lessons in all subjects to suit and cater to students that were, blind, hospital bound, mute, autistic, and the like.

Social Work/ Crime Victim Advocate | *Safe Horizon / NYPD* | 2018 – 2020

- **High-Stakes Coordination:** Managed a caseload of 100+ clients/week, coordinating with law enforcement and the criminal justice system to provide trauma-informed interventions as well as connections to social resources.
- **Program Orchestration:** Led monthly community outreach and educational events to facilitate positive relations between the public and mental health services. Reached out and coordinated with local businesses and leaders of the community to spread awareness of resources available to those in need.
- **Client Counseling:** Engaged in safety planning and client centered practice to provide solutions and connect individuals to resources across different organizations. Handled high intensity cases with modesty and understanding, giving support as needed to clients.

Physician Research Assistant | *Bellevue Hospital* | 2017 – 2018

- **Statistical Analysis:** Conducted survey analysis on thousands of patient records to identify undiagnosed clinical depression, utilizing R for data visualization (Histograms, Pie Charts, Plot Graphics).
- **Literary Reviews:** Composed literary reviews of Asian cultures to better identify and address stigma concerning mental health interventions. Reviewed findings with licensed clinicians and assisted in developing public health interventions to address documented issues.

EDUCATION & CERTIFICATIONS

- **Bachelor of Science: Public Health** | Monroe College (GPA: 3.94)
- **Certifications:** Google Data Analytics Certificate, Coding Fundamentals I & II, Intro to Data Science, Modern Project Management.
- **Languages:** Intermediate Japanese.